

Diagnosing Separation and Initial-Condition Errors

Answer Key

AP Calculus AB/BC • Error Analysis

Quick Answers

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|------------------|-----------|---------|
| 1. $2x^4 - 3x^2$ | 4. 890.48 | 7. 6.83 |
| 2. 25 | 5. 5 | 8. — |
| 3. 8.15 | 6. 4 | |
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Curriculum

Level: AP Calculus AB/BC • Error Analysis

FUN-7 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 8 tagged checks.

Common wrong answers

2. If they got 20: dropped the constant of integration before evaluating the particular solution
 3. If they got 4: integrated the right side as though y were a constant, instead of separating the variables first
 4. If they got 153.41: added the constant after exponentiating instead of carrying it through as a multiplying factor
 5. If they got 7: took the square root of each term separately, turning a sum of squares into a sum
 6. If they got 1.33: lost the minus sign from the integral of y to the negative second power
 7. If they got 4.83: left the constant of integration out, reporting the antiderivative rather than the particular solution
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1. General solution of $\frac{dy}{dx} = 8x^3 - 6x$.

Step 1 (check for separation). The right side holds no y , so the variables are already separated and the equation integrates directly. **Step 2 (integrate).** $y = \int (8x^3 - 6x) dx$,

so

$$y = 2x^4 - 3x^2 + C$$

Step 3 (interpret C). On a slope field, C selects which member of the family of parallel solution curves you are on — every curve has the same slope at a given x , so they differ only by a vertical shift.

2. Same equation with $y(1) = 4$; evaluate $y(2)$.

Step 1 (integrate first). The general solution is $y = 2x^4 - 3x^2 + C$. The initial condition can be used only now, once a C exists to solve for — that is the step it belongs to. **Step 2 (apply the condition).** $y(1) = 2 - 3 + C = 4$, so $C = 5$ and the particular solution is $y = 2x^4 - 3x^2 + 5$. **Step 3 (evaluate).** $y(2) = 32 - 12 + 5 = \boxed{25}$. A student who integrates and then evaluates without solving for C reports 20 — the antiderivative, not the solution of this initial-value problem.

3. Find and fix: $\frac{dy}{dx} = 2xy$, $y(0) = 3$, find $y(1)$.

Step 1 (name the first wrong line). The very first line. Wren writes $\int dy = \int 2xy dx$ and then integrates the right side as if y were a constant. It is not: y is the unknown *function* of x , so $\int 2xy dx$ cannot be evaluated at all until the y is moved to the other side.

Step 2 (separate, then integrate). $\frac{dy}{y} = 2x dx$, so $\ln |y| = x^2 + C$ and $y = Ae^{x^2}$ with $A = e^C$. **Step 3 (apply the condition and evaluate).** $y(0) = A = 3$, so $y = 3e^{x^2}$ and $y(1) = 3e \approx \boxed{8.15}$. Check: $y' = 3e^{x^2}(2x) = 2xy \checkmark$. Wren's 4 is the answer to a different, non-separable computation.

4. Find and fix: $\frac{dy}{dx} = 5y$, $y(0) = 6$, find $y(1)$.

Step 1 (name the first wrong line). The second. $\ln |y| = 5x + C$ is correct; exponentiating it is not $y = e^{5x} + C$. Exponentiating a *sum* gives a *product*: $e^{5x+C} = e^C e^{5x}$. **Step 2 (carry the constant as a factor).** Write $A = e^C$, so the general solution is $y = Ae^{5x}$ — the constant multiplies, it does not add. **Step 3 (apply the condition and evaluate).** $y(0) = A = 6$, so $y = 6e^{5x}$ and $y(1) = 6e^5 \approx \boxed{890.48}$. Check: $y' = 30e^{5x} = 5y \checkmark$. Dev's 153.41 fails that check, which is the one-line test that would have caught the error before any arithmetic.

5. Find and fix: $\frac{dy}{dx} = \frac{x}{y}$, $y(0) = 4$, find $y(3)$.

Step 1 (name what is wrong). The square root was taken term by term. $\sqrt{u^2 + v^2}$ is not $u + v$; the radical must be applied to the whole side at once, after the constant has been folded in. **Step 2 (do it correctly).** From $\frac{y^2}{2} = \frac{x^2}{2} + C$, multiply through by 2 to get $y^2 = x^2 + K$ with $K = 2C$. Applying $y(0) = 4$ gives $16 = 0 + K$, so $K = 16$ and $y = \sqrt{x^2 + 16}$ (the positive root, because $y(0)$ is positive). **Step 3 (evaluate).** $y(3) = \sqrt{9 + 16} = \boxed{5}$. Check: $y' = \frac{x}{\sqrt{x^2 + 16}} = \frac{x}{y} \checkmark$. Rosa's 7 is exactly $3 + 4$, the signature of the term-by-term root.

6. Find and fix: $\frac{dy}{dx} = y^2$, $y(0) = 2$, find $y(0.25)$.

Step 1 (name the missing sign). $\int y^{-2} dy = -\frac{1}{y}$, not $+\frac{1}{y}$. The power rule sends y^{-2} to $\frac{y^{-1}}{-1}$, and Sam dropped that minus. **Step 2 (redo with the sign).** $-\frac{1}{y} = x + C$, and $y(0) = 2$ gives $-0.5 = C$. So $-\frac{1}{y} = x - 0.5$, that is $y = \frac{1}{0.5 - x}$. **Step 3 (evaluate and read the behaviour).** $y(0.25) = \frac{1}{0.25} = \boxed{4}$. As $x \rightarrow 0.5^-$ the denominator goes to zero and $y \rightarrow \infty$: the solution blows up in finite time, so its interval of existence stops at $x = 0.5$. Sam's version, $y = \frac{1}{x + 0.5}$, decays smoothly forever and hides that blow-up entirely — the sign error changes the qualitative behaviour, not just a digit.

7. $\frac{dy}{dx} = \frac{6x}{x^2 + 1}$, $y(0) = 2$; evaluate $y(2)$.

Step 1 (integrate). With $u = x^2 + 1$ the right side integrates to $3 \ln(x^2 + 1)$, so $y = 3 \ln(x^2 + 1) + C$. **Step 2 (apply the condition).** $y(0) = 3 \ln 1 + C = C = 2$, so $y = 3 \ln(x^2 + 1) + 2$. Applying the condition *before* integrating would be meaningless: there is no constant to determine yet, and $\frac{dy}{dx}$ is a statement about a slope, not about a value of y . **Step 3 (evaluate).** $y(2) = 3 \ln 5 + 2 \approx \boxed{6.83}$. A student who never writes the constant reports 4.83 — the antiderivative evaluated at 2, which satisfies the differential equation but not the initial condition.

8. Refute: “a constant may be added before or after exponentiating.” *Open reasoning — read by hand.*

Model argument. Differentiate the claimed solution: if $y = e^{5x} + 5$ then $y' = 5e^{5x}$, while $5y = 5e^{5x} + 25$. These differ by 25 for every x , so the claimed function does not satisfy $\frac{dy}{dx} = 5y$ and the placement of the constant is not a matter of taste. **What the constant becomes.** Integrating gives $\ln |y| = 5x + C$; exponentiating turns the sum into a product, $y = e^C e^{5x} = A e^{5x}$. So the additive constant of integration becomes a *multiplicative* factor A , and A , not C , is what the initial condition determines. **Accept:** any argument that produces the mismatch by differentiation, or that shows $y = e^{5x} + 5$ fails the equation at a single specific x (for example $x = 0$: $y' = 5$ but $5y = 30$). **Do not accept:** “because the book says $A e^{5x}$ ” with no test, or a claim that both forms work on some interval.